CUSTOMER CHURN PREDICTION

SQL Analysis Report

Database Queries · Results · Key Insights

Metric	Value
Total Customers	**999,999**
Churned Customers	**99,227 (9.92%)**
Retained Customers	**900,772 (90.08%)**
Database	Microsoft SQL Server 16.0
Table Name	dbo.customers
Total Columns	38 columns
Queries Executed	5 analytical queries

Role: SQL Engineer | Database: Microsoft SQL Server | Table: customers | Rows: 999,999

1. Database Overview

The customer churn data was loaded from a preprocessed CSV file into Microsoft SQL Server. The database customer_churn contains one table — dbo.customers — holding 999,999 rows and 38 columns after one-hot encoding of categorical features during the Data Engineering phase.

Property	Detail
Server	OMAR-ZIZO (SQL Server 16.0.1180)
Database	customer_churn
Schema	dbo
Table	customers
Primary Key	customer_id (VARCHAR 50)
Row Count	**999,999**
Column Count	38
Data Types	FLOAT, INT, VARCHAR, DATE

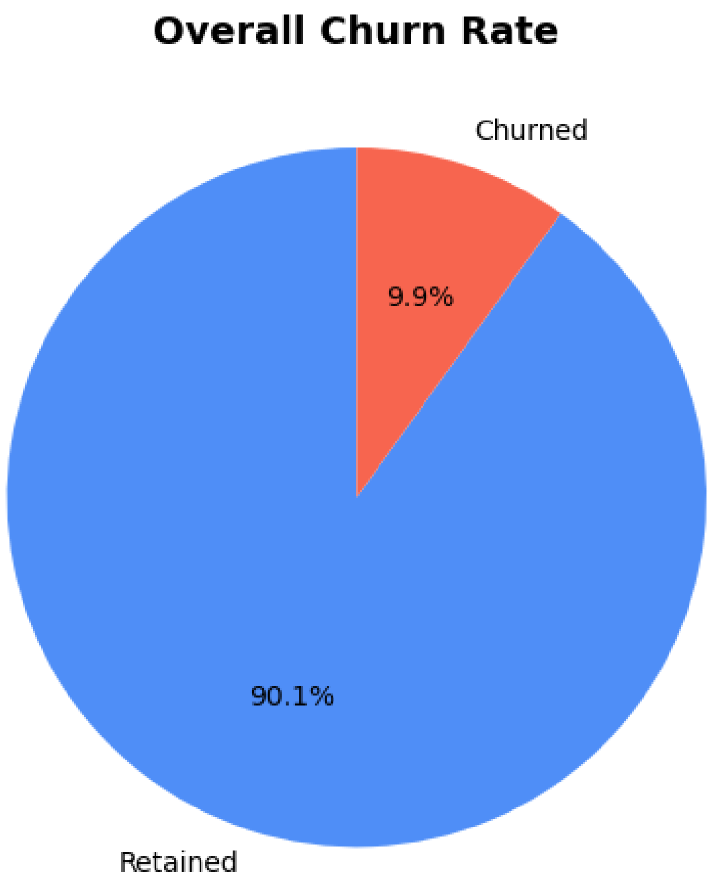
2. Query 1 — Overall Churn Rate

The first query calculates the total number of customers, how many churned, and the overall churn rate across the entire dataset.

```
SELECT
    COUNT(*)                                AS total_customers,
    SUM(churn)                              AS churned,
    ROUND(AVG(CAST(churn AS FLOAT))*100,2) AS churn_rate
FROM customers;
```

total_customers	churned	churn_rate
999,999	**99,227**	**9.92%**

> **Class Imbalance of 9:1 — only 9.92% of customers churned. Accuracy alone is misleading; F1-Score and Recall are more meaningful metrics.**



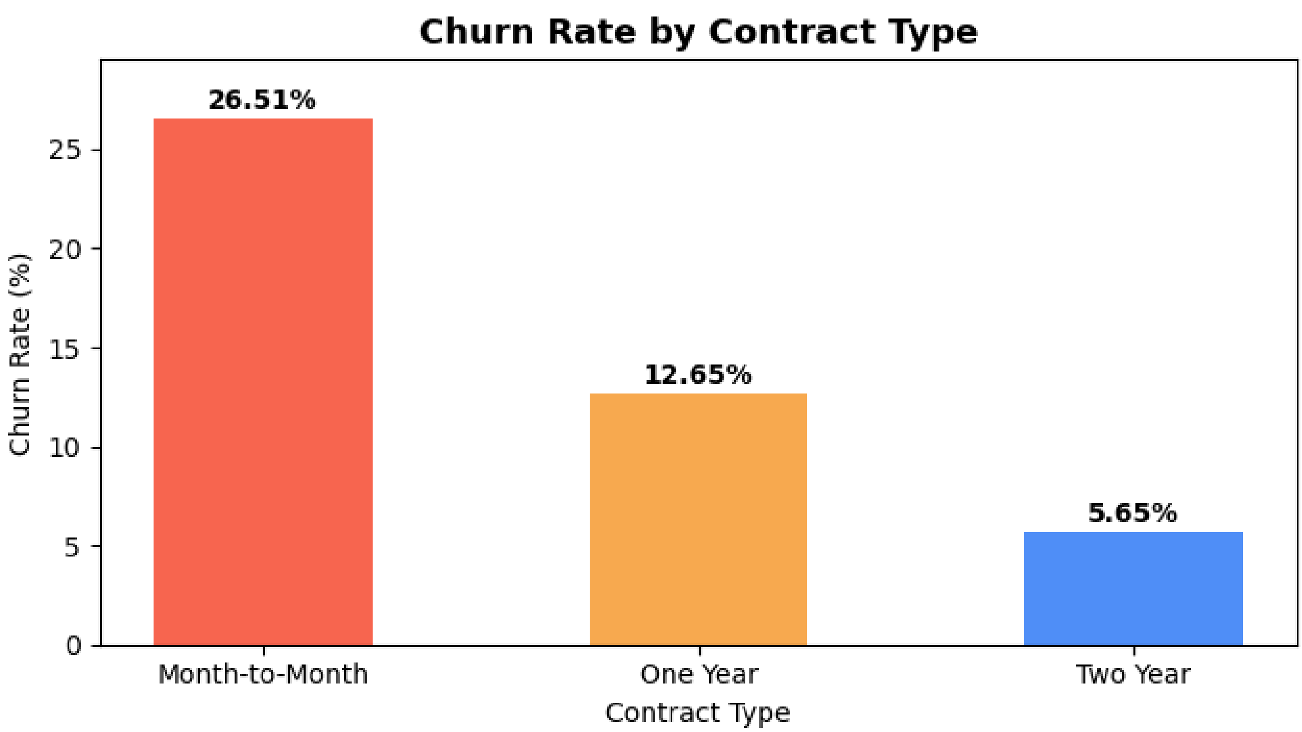
3. Query 2 — Churn by Contract Type

Contract type is one of the strongest predictors of churn. This query groups customers by their contract type and calculates the churn rate for each group.

```
SELECT
    contract,
    COUNT(*)                                AS total,
    SUM(churn)                              AS churned,
    ROUND(AVG(CAST(churn AS FLOAT))*100,2) AS churn_rate
FROM customers
GROUP BY contract
ORDER BY churn_rate DESC;
```

Contract	Total	Churned	Churn Rate
0 — Month-to-Month	19,992	5,299	**26.51%**
1 — One Year	550,468	69,651	**12.65%**
2 — Two Year	429,539	24,277	**5.65%**

> **Month-to-Month customers churn at 26.51% — nearly 5x the rate of Two-Year contract holders (5.65%). Encouraging customers to switch to longer contracts is a high-impact retention strategy.**



4. Query 3 — High-Risk Customers

This query identifies the top 100 customers who are at highest risk of churning. They are currently retained (churn = 0) but show warning signals: high complaint counts and low satisfaction scores.

```
SELECT TOP 100
    customer_id, num_complaints,
    customer_satisfaction, contract, churn
FROM customers
WHERE num_complaints > 0
  AND customer_satisfaction < 0
  AND churn = 0
ORDER BY num_complaints DESC;
```

Sample Results (Top 10 of 100 rows):

customer_id	num_complaints	customer_satisfaction	contract	churn
160873	**7.59**	-0.08	2	0
746840	**7.59**	-0.51	1	0
821278	**7.59**	-1.81	2	0
940033	**7.59**	-0.94	1	0
20993	**6.39**	-1.81	2	0
31848	**6.39**	-0.94	2	0
88451	**6.39**	-0.94	2	0
111210	**6.39**	-0.51	2	0
151474	**6.39**	-0.08	1	0
170845	**6.39**	-0.08	1	0

> **These 100 customers are currently retained but show the highest complaint scores and lowest satisfaction — they are the most likely to churn next. Proactive outreach and service recovery should be prioritized for this group.**

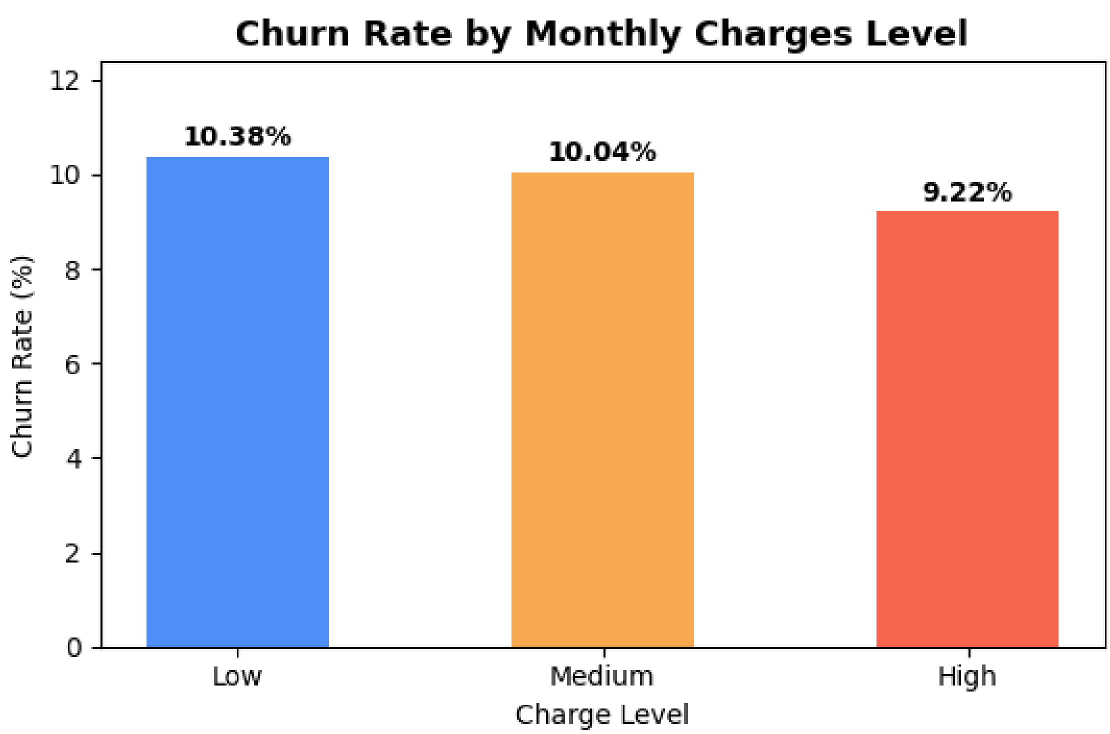
5. Query 4 — Churn by Monthly Charges

Customers were grouped into Low, Medium, and High billing tiers based on their scaled monthly charges. Churn rate was calculated for each tier.

```
SELECT
    CASE
        WHEN monthlycharges < -0.5 THEN 'Low'
        WHEN monthlycharges <  0.5 THEN 'Medium'
        ELSE 'High'
    END AS charge_level,
    ROUND(AVG(CAST(churn AS FLOAT))*100,2) AS churn_rate
FROM customers
GROUP BY CASE
        WHEN monthlycharges < -0.5 THEN 'Low'
        WHEN monthlycharges <  0.5 THEN 'Medium'
        ELSE 'High'
    END;
```

Charge Level	Churn Rate
Low	10.38%
Medium	10.04%
High	9.22%

> **Monthly charge level has minimal impact on churn — rates are nearly equal across Low (10.38%), Medium (10.04%), and High (9.22%). This suggests pricing alone is not a primary churn driver.**



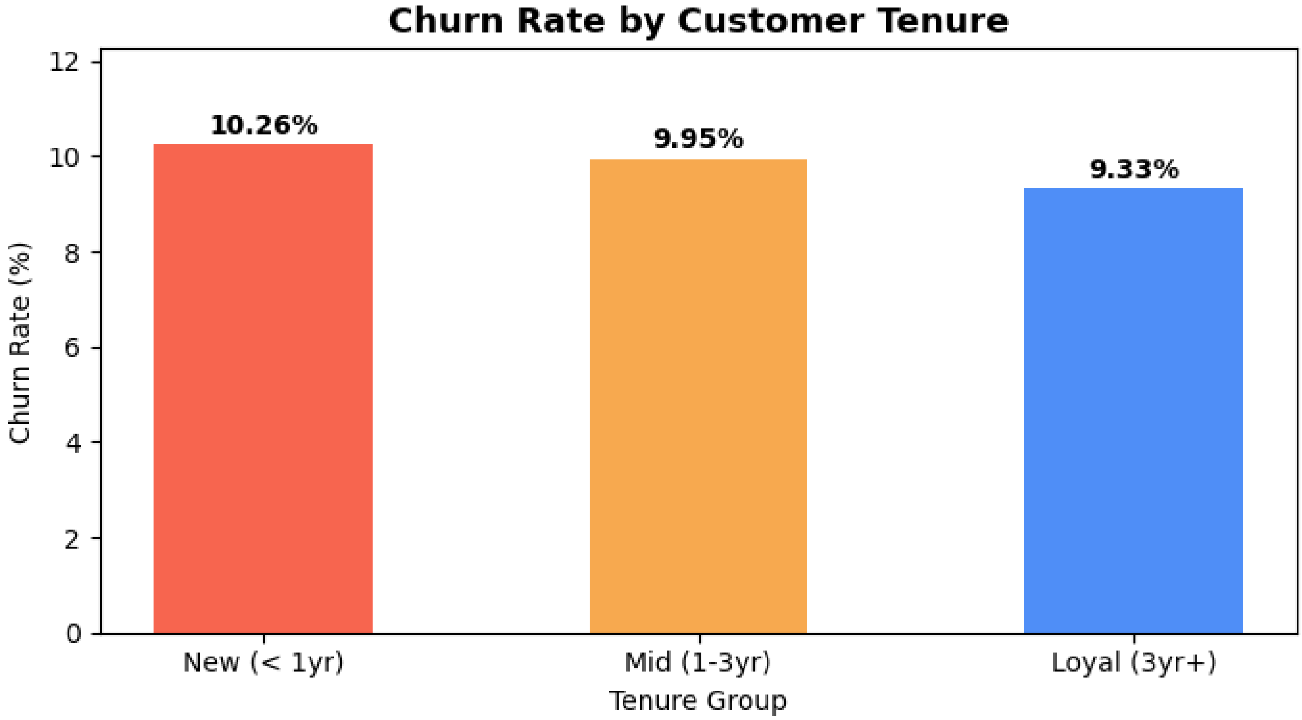
6. Query 5 — Churn by Tenure

Customers were grouped into three tenure bands to understand how customer lifetime affects churn probability.

```
SELECT
    CASE
        WHEN tenure < -0.5 THEN 'New (< 1yr)'
        WHEN tenure <  0.5 THEN 'Mid (1-3yr)'
        ELSE 'Loyal (3yr+)'
    END AS tenure_group,
    COUNT(*)                               AS total,
    SUM(churn)                             AS churned,
    ROUND(AVG(CAST(churn AS FLOAT))*100,2) AS churn_rate
FROM customers
GROUP BY CASE ... END
ORDER BY churn_rate DESC;
```

Tenure Group	Total	Churned	Churn Rate
New (< 1yr)	418,054	42,886	**10.26%**
Mid (1-3yr)	329,206	32,757	9.95%
Loyal (3yr+)	252,739	23,584	**9.33%**

> **New customers (< 1 year) have the highest churn rate at 10.26%. Loyal customers (3yr+) are the most stable at 9.33%. Early customer engagement programs in the first year could significantly reduce churn.**



7. Key Insights Summary

Based on the five SQL queries executed against the customer_churn database, the following insights were identified:

#	Insight	Business Action
1	Overall churn rate is 9.92%	*Use F1-Score & Recall — not accuracy*
2	Month-to-Month contracts churn at 26.51%	*Offer discounts to switch to annual plans*
3	Two-Year contracts churn at only 5.65%	*Promote long-term contracts aggressively*
4	100 high-risk customers identified	*Proactive outreach & service recovery*
5	Monthly charges have minimal churn impact	*Focus retention on contract type instead*
6	New customers (< 1yr) churn most at 10.26%	*Launch onboarding & early loyalty programs*

8. SQL Analysis Pipeline Summary

Step	Action	Result
1	**Load CSV to SQL Server**	999,999 rows inserted via pyodbc
2	**Query — Overall Churn Rate**	9.92% overall churn
3	**Query — Churn by Contract**	Month-to-Month = 26.51% churn
4	**Query — High-Risk Customers**	100 at-risk customers identified
5	**Query — Churn by Charges**	Minimal variation across tiers
6	**Query — Churn by Tenure**	New customers churn most (10.26%)

End of SQL Analysis Report